

Pentacene–Bridge Interactions in an Axially Chiral Binaphthyl Pentacene Dimer

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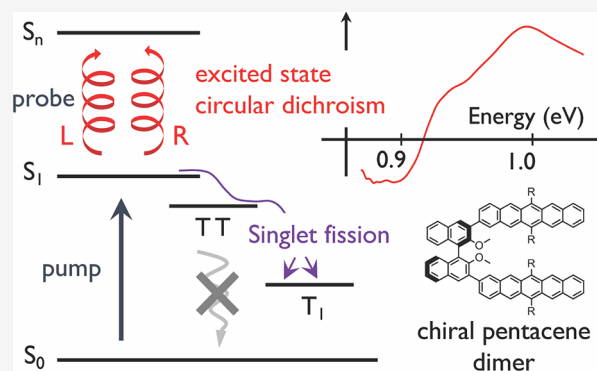


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ABSTRACT: Molecular chirality can be exploited as a sensitive reporter of the nature of intra- and interchromophore interactions in π -conjugated systems. In this report, we designed an intramolecular singlet fission (iSF)-based pentacene dimer with an axially chiral binaphthyl bridge (2,2'-(2,2'-dimethoxy-[1,1'-binaphthalene]-3,3'-diyl) *n*-octyl-di-isopropyl silyl ethynyl dipentacene, **BNBP**) to utilize its chiroptical response as a marker of iSF chromophore–bridge–chromophore (SFC– β –SFC) interactions. The axial chirality of the bridge enforces significant one-handed excitonic coupling of the pentacene monomer units; as such, **BNBP** exhibits significant chiroptical response in the ground and excited states. We analyzed the chiroptical response of **BNBP** using the exciton coupling method and quadratic response density functional theory calculations to reveal that higher energy singlet transitions in **BNBP** involve significant delocalization of the electronic density on the bridging binaphthyl group. Our results highlight the promising application of chiroptical techniques to investigate the nature of SFC– β –SFC interactions that impact singlet fission dynamics.



INTRODUCTION

Singlet fission (SF),¹ the spin-allowed evolution of a singlet exciton into two triplet excitons, has promising potential in the field of photovoltaics.^{2,3} Thermalization of photon energy in excess of the absorbing threshold, which constitutes a major loss pathway in solar cells, can be significantly reduced in a SF-based solar cell. The ability of SF to increase the efficiency of silicon solar cells is of particular interest and has been addressed theoretically⁴ and recently demonstrated.^{5,6} Detailed insights into the SF process have been developed in the past decade, largely enabled by investigations of the excited-state dynamics of covalently linked SF-based chromophores in dilute solutions.^{7–18} Here, SF is an intramolecular process, referred to as intramolecular singlet fission (iSF), that can be controlled by the molecular structure of the bridge. In such systems, parameters such as the size,¹⁶ conjugation,^{13,19} energetics,^{14,20} conformation,^{21–23} and linking position^{24,25} of the bridge can be used to modulate coupling between monomers and consequently the iSF dynamics. In addition to bridge-mediated coupling between monomers, monomer–bridge interactions are anticipated to be crucial to the iSF process,^{24,26–29} but insights into the latter are limited. Understanding this aspect is crucial to rational design of efficient SF-based chromophores, necessitating the design of molecular systems where monomer–bridge interactions can be

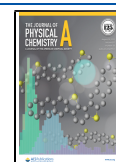
selectively investigated and the use of a diversity of optical probes sensitive to monomer–bridge interactions.

In covalently bridged dimers where the bridge is chiral, its chiroptical response, i.e., a result of its dissymmetric interaction with circularly polarized light, can be implemented to probe intramolecular interactions.^{30–33} For instance, circular dichroism (CD), which measures the difference in the absorption of left (L) and right (R) handed circularly polarized light, has been often used to determine the absolute ground-state configuration of dimers.^{34–36} The CD signal of chiral dimers is remarkably sensitive to coupling between monomers and is of a much longer range ($\sim 1/R_{12}^2$) than the dipolar intermolecular interaction, which scales $1/R_{12}^3$, where R_{12} is the vector separation between the monomers (or molecules).^{37,38} As such, the CD signatures of chiral covalently bridged dimers have previously been utilized to investigate excitonic coupling between monomer transitions,³⁶ perturbations of the monomer transitions due to the chiral bridge,^{39,40} coupling of

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electronic and magnetic transition dipole moments in the dimer,^{41,42} and wave function overlap between the monomer and the chiral bridge units. While CD is responsive to molecular arrangements in dimers at the ground state, excited-state CD is highly sensitive to coupling between monomers and monomer–bridge interactions in the excited state. Recent work on excited-state CD in axially chiral dimers⁴³ and cholesteric phases of conjugated polymers^{44,45} has revealed crucial insights into the role of molecular structure and its rearrangement on the excited-state relaxation dynamics. However, little is known about the impact of chirality on iSF dynamics. In this vein, a combination of CD and excited-state CD techniques can be used to sensitively investigate the role of SFC- β -SFC interactions in systems containing axial chirality.

In this report, we incorporate an axially chiral binaphthyl unit as a bridge between two pentacenes to design a chiral pentacene dimer (**BNBP**, Figure 1) which undergoes iSF and

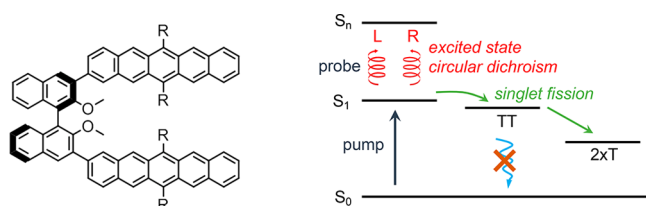


Figure 1. (Left) Molecular structure of the axially chiral **BNBP** pentacene dimer. R denotes the *n*-octyl-di-isopropyl silyl ethynyl (NODIPS) group. (Right) Schematic of the excited-state circular dichroism and singlet fission phenomena. L and R are left and right circularly polarized light, respectively. TT and 2xT represent coupled and uncoupled triplets, respectively.

also exhibits significant CD and excited-state CD responses. The iSF dynamics in **BNBP** involve the formation of a very long-lived iSF intermediate triplet pair (TT) state, consistent with previous reports on iSF dynamics of covalently bridged pentacene dimers with homoconjugated bridges.¹³ Interestingly, the triplet–triplet recombination in the TT state in **BNBP** is significantly suppressed, which we attribute to the weak pentacene–pentacene coupling in **BNBP** and pentacene–bridge interactions. Analysis of the CD and excited-state CD responses reveals a significant contribution from the binaphthyl unit to the high-energy singlet transitions in **BNBP**. This is further confirmed using *ab initio* linear response time-dependent density functional theory (TD-DFT) and quadratic response (QR-DFT) calculations, which suggest that high-energy singlet transitions in **BNBP** involve significant delocalization of the electronic density on the bridging binaphthyl group. It is likely that this wave function overlap between the pentacene units and the binaphthyl bridge aids in the significant suppression of the triplet–triplet recombination in **BNBP**, in agreement with recent reports^{20,26,27} on the role of the frontier molecular orbitals of the bridge on the decay dynamics of the TT state. Our approach highlights the promising role of chiroptical techniques such as CD and excited-state CD to sensitively investigate SFC- β -SFC interactions and the impact on multiexciton dynamics.

RESULTS AND DISCUSSION

Similar to previous reports on iSF in pentacene dimers coupled through the 2,2' position,^{13,16} we use complementary time-resolved electron spin resonance (tr-ESR) and transient absorbance (TA) techniques to establish the exciton dynamics

in **BNBP**. In Figure 2, continuous wave (cw) and pulsed tr-ESR measurements are presented. In Figure 2a, an initially

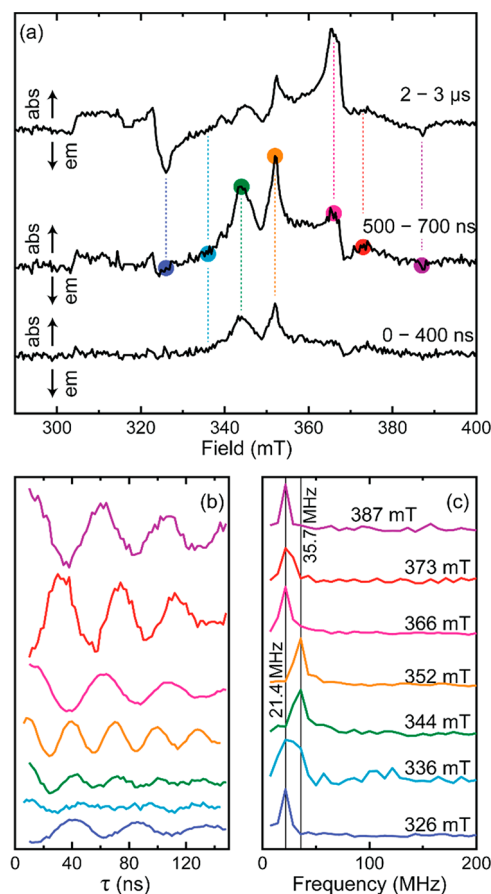


Figure 2. ESR measurements of **BNBP** in toluene at 80 K after laser excitation at 590 nm. (a) Pulsed laser, cw-microwave tr-ESR at a microwave frequency of 9.701984 GHz integrated over three temporal ranges. Positions of nutation experiments shown in b are marked with colored circles. (b) Echo signals from nutation experiments with a pulse sequence τ – π –echo. (c) Fourier transformation of the data in b.

narrow spectrum (0–400 ns) evolves into a broader spectrum (2–3 μ s), similar to the generation of TT pairs evolving into decoupled triplets seen in previously reported singlet fission materials.^{12,13,17,46–49} The Landé *g* factor and zero-field splitting *D* parameter of the dominant species at 2–3 μ s are ~ 2 and 1.1 GHz, respectively, in good agreement with previous observations of uncoupled pentacene triplets. To confirm this assignment, Rabi nutation experiments (Figure 2b and 2c) were carried out to establish the spin multiplicities of the observed states. Features associated with the initially generated spectrum (0–400 ns) have Rabi nutation frequencies of 35.7 MHz, as compared to 21.4 MHz for the features in the 2–3 μ s spectrum. The nutation frequency ratio of $1.67 \approx \sqrt{3}$ is the expected ratio between strongly and weakly coupled triplets generated by singlet fission (where the strongly coupled triplets occupy the spin quintet with zero projection upon the magnetic field direction). Although tr-ESR measurements confirm that **BNBP** undergoes iSF, they can only measure population differences between sublevels of a multiplet. Therefore, techniques such as TA spectroscopy are

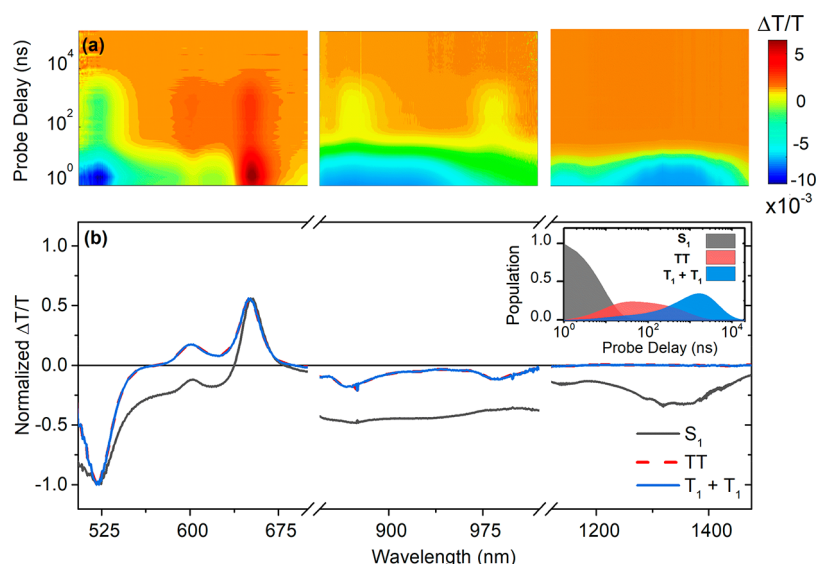


Figure 3. (a) Differential changes in the transmission intensity (visible and near-infrared) measured in nanosecond (ns) transient absorption (TA) measurements of **BNBP** in dichloromethane (DCM) at room temperature. Pump excitation wavelength was 400 nm ($60 \mu\text{J}/\text{cm}^2$). (b) Normalized deconvoluted evolution-associated transient absorption spectra of the singlet state (S_1), the triplet pair state (TT), and the uncoupled triplets ($T_1 + T_1$) for **BNBP** solution in DCM, as obtained from global fitting using a sequential model. TA spectrum of uncoupled triplets is identical to the spectrum of the TT state (see SI Note S2 for details on fitting). (Inset in b) Normalized species populations obtained from the global fit of **BNBP** solutions in DCM.

required to reveal insights into the dynamics and efficiency of iSF in **BNBP**.

We probed the TA signatures in the spectral window spanning the visible and near-infrared (NIR) region (Figure 3a, Figures S2–S5). The shape of the early time (<10 ns) TA spectra of **BNBP** exhibits features characteristic of singlet absorption in pentacenes,¹⁸ with photoinduced absorption (PIA) peaks at 522 and 1380 nm and a ground-state bleach (GSB) peak at 655 nm (subnanosecond data is shown in the SI). These features evolve over tens of nanoseconds into signatures reminiscent of long-lived triplet species in pentacene dimers.¹⁶ Notably, the shape of the TA spectra at longer probe delays (greater than hundreds of nanoseconds) in the region ($800 < \lambda < 1000$ nm) are considerably narrower in contrast to the broad, overlapping singlet photoinduced absorption (PIA) features at early time scales.

To extract the spectral signature and time evolution of different species in the TA data, we employ a global fit approach. We note that similar to the PIA of weakly coupled pentacene dimers reported previously, triplet PIA features in **BNBP** exhibit biexponential decay kinetics without any change in the shape of the spectra.¹³ However, to describe accurately the dynamics of the system, we require the inclusion of the triplet pair (TT) and the uncoupled triplet ($T_1 + T_1$). Further details of the fitting approach are given in Note S2, which describes the deconvolution of species in the TA data of **BNBP**.

The short-lived species is confirmed as the photoexcited S_1 state, which decays with a monoexponential decay component of about 8.9 ns (Figures S2–S5, Table S1). In addition to the initially excited S_1 state, intermediate (~ 900 ns) and a long-lived species (~ 4 – $6 \mu\text{s}$) are necessary to fit the TA data; their spectral shapes accord with the previously reported spectral signatures of the triplet absorption spectrum in pentacenes¹⁸ and bipentacenes.^{13,16,17} Since the tr-ESR showed definitively the existence of strongly coupled triplet pairs and uncoupled triplets, the intermediate and long-lived species are assigned to

the TT and uncoupled triplet ($T_1 + T_1$) states, respectively. As expected, the GSB of the TT spectrum is identical to the combined GSB of the corresponding uncoupled triplets. The spectra of TT and uncoupled triplets in **BNBP** are strikingly similar, suggesting weak exchange coupling between the individual triplets in the triplet pair state.⁵⁰

The inset in Figure 3b shows the time evolution of the singlet, TT, and ($T_1 + T_1$) state populations for **BNBP** in DCM. The global fitting of the TA data yields quantum efficiencies of singlet fission (ϕ_{sf}), intersystem crossing (ϕ_{isc}), and triplet pair decoupling (ϕ_{sep}) values of 0.22, 0.05, and 1, respectively, with the corresponding lifetimes of S_1 , TT, and ($T_1 + T_1$) of 8.9, 936, and 3770 ns, respectively. One of the main results from the extracted time evolution of the species is the value of ϕ_{sep} in **BNBP**. The fit for the TA data of **BNBP** only simulates the data accurately for $\phi_{\text{sep}} > 0.85$, and the best fit of the data is obtained for a ϕ_{sep} value of 1 (for details, see Note S2, Figures S6–S9). The near unity value of ϕ_{sep} for **BNBP** suggests quantitative decoupling of the TT states into uncoupled triplet states. In other words, triplet–triplet recombination of the TT state to the ground state is significantly suppressed in **BNBP**.

Comparing the iSF dynamics of **BNBP** to other similar pentacene dimer systems reported in the literature,^{8,11–13,16,17,19,25,51} it can be observed that both the rate of TT formation as well as the rate of triplet–triplet recombination in the TT state in **BNBP** are significantly low (see Table S2, Figures S10–S13). As iSF in these systems has been shown to be very sensitive to the nature of interpentacene coupling as well as pentacene–bridge interactions, further insights into the nature of electronic coupling in **BNBP** are necessary. To this end, we employ CD and excited-state CD techniques to sensitively investigate the electronic structure of **BNBP** in the ground and excited states, respectively.

We first compare the ground-state absorption signature of **BNBP** and the monomer pentacene (Pc) to identify the origins of transitions in **BNBP** (Figure 4). The shape of the

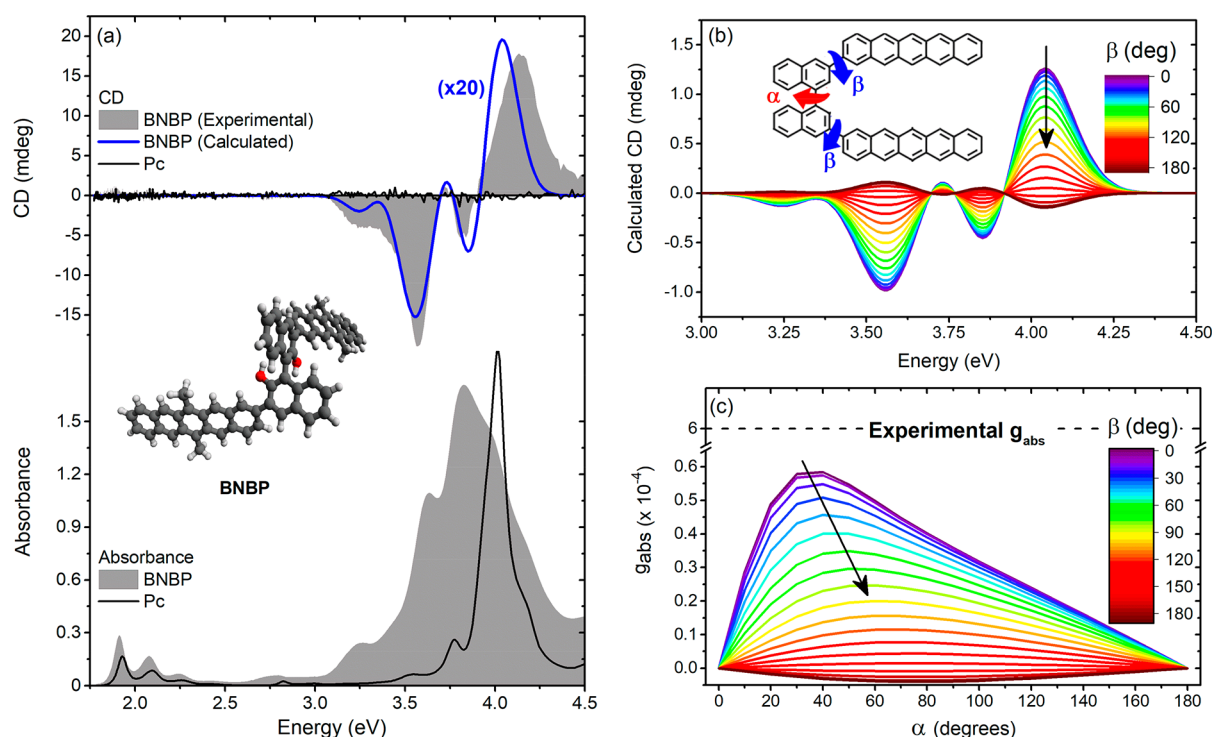


Figure 4. (a) Measured CD (upper panel), calculated CD (upper panel), and absorption (lower panel) spectra of 10 μM solutions of **BNBP** and the monomer pentacene (**Pc**) dissolved in DCM at room temperature. (Inset in a) Ground-state optimized geometry of **BNBP** obtained through DFT. (b) Calculated CD spectra of **BNBP** for different values of β , for a constant value of α of 90° . (c) Calculated values of g_{abs} of **BNBP** at 4.0 eV for different values of α and β . (Inset in b) Dihedral and torsion angles, marked as α and β , respectively. In c, dashed line denotes the experimental value of g_{abs} of **BNBP** at 4.0 eV.

absorption spectra of **BNBP** and **Pc** is very similar in the visible region, indicating that the origins of the low-energy optical transitions in **BNBP** and **Pc** are essentially the same. Thus, the low-energy absorption peaks around 525–650 nm in **BNBP** can be attributed to the vibrational manifold of the first excited singlet transition, polarized along the short axis of the monomer.⁵² Conversely, intense transitions around 310 nm are predominantly polarized along the long axis of the monomer. In contrast to the similarity in the absorption spectra of **BNBP** and **Pc** in the visible region, the long-axis polarized transitions of **BNBP** in the UV region are significantly broadened. The observed broadening of the high energy absorption peaks of **BNBP** can arise due to excitonic coupling of the long-axis polarized transitions whereby the dipole moments of individual pentacene units combine in-phase and out-of-phase, giving two distinct absorptions in the dimer, as well as due to the emergence of new transitions resulting from orbital interactions between the pentacene units and the bridging unit.⁵³

As the bridge in **BNBP** is axially chiral, CD measurements can be used to sensitively investigate the extent of excitonic coupling and the molecular arrangement of the pentacene moieties in the ground state. Figure 4a shows the measured room-temperature CD of isolated **BNBP** molecules and monomer pentacene molecules in DCM. The CD signal is conveniently reported in units of millidegrees, which can be directly related to the difference in absorption of L and R light (eq 1)³⁰

$$\text{CD (mdeg)} = 32\,982(A_L - A_R) \quad (1)$$

where A_L and A_R represent the absorbance of left- and right-handed circularly polarized light, respectively. In contrast to the absence of any CD in solutions of **Pc**, **BNBP** solutions

exhibit a significant bisignate CD response, suggesting one-handed excitonic coupling of the pentacene units in **BNBP**.³⁶ Further, while a bisignate CD feature is observed for the long-axis transitions (250–400 nm), the short-axis transitions (525–650 nm) do not show CD; this implies a near orthogonal arrangement³¹ of the planes containing the two pentacene chromophores. This is also confirmed by calculating the ground-state-optimized geometry of **BNBP** through density functional theory (DFT) calculations, revealing that the dihedral angle between the pentacene units in **BNBP** is around 78° (inset in Figure 4a).

Assuming the origin of the CD in **BNBP** is purely due to excitonic coupling of monomeric transitions, the CD spectrum of **BNBP** can be calculated through the excitonic coupling method (for details, see Note S3, Table S3).⁵⁴ Briefly, the CD spectrum of the dimer is determined by calculating the rotational strengths (R_+ and R_-) and electronic coupling (V) of the excitonically coupled dimer states given that the molecular geometry and the dipole moment vectors (μ_1 and μ_2) of the monomeric transitions are known (eqs 2 and 3).

$$R_{\pm} = \mp \pi \omega_0 [\mathbf{R}_{12} \cdot (\mu_1 \times \mu_2)] \quad (2)$$

$$V = \frac{1}{R_{12}^3} \left[(\mu_1 \cdot \mu_2) - \frac{3(\mu_1 \cdot \mathbf{R}_{12})(\mu_2 \cdot \mathbf{R}_{12})}{R_{12}^2} \right] \quad (3)$$

Here, $\mathbf{R}_{12}$ is the intermonomer separation vector and ω_0 is the resonant frequency of the monomeric transition.

Figure 4a shows the calculated CD spectrum of **BNBP** considering the molecular geometry is ground-state optimized (inset in Figure 4a) and the monomeric transitions responsible for CD are polarized along the long axis of the pentacene units.

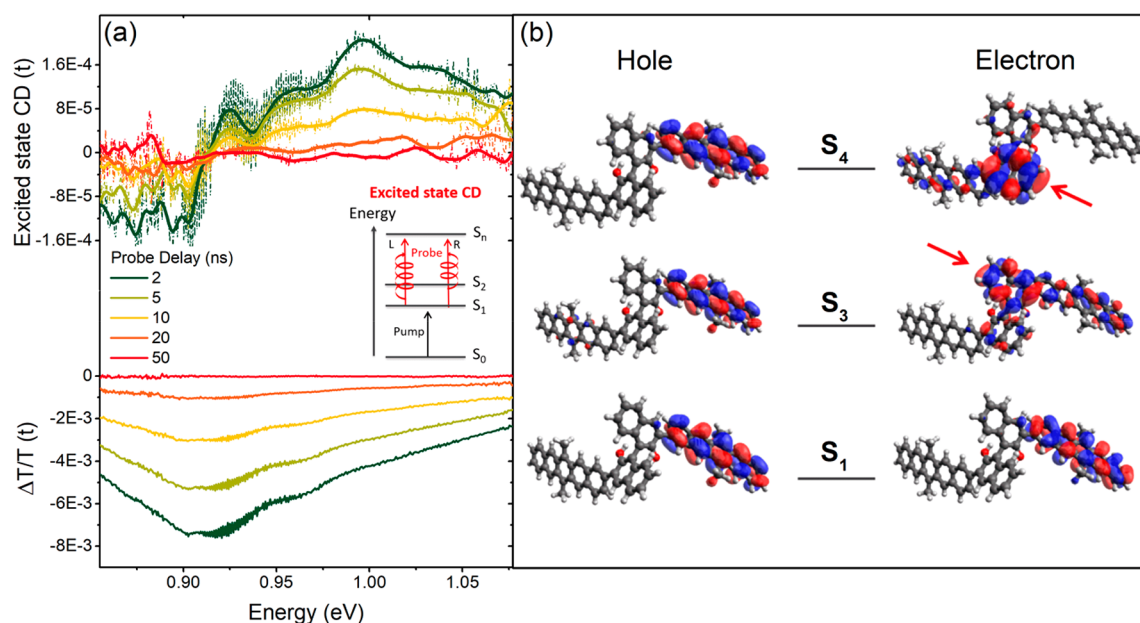


Figure 5. (a) Measured excited-state CD (upper panel) and differential changes in the transmission intensity ($\Delta T/T$) spectra of 10 μM solutions of BNPB dissolved in DCM at room temperature as a function of nanosecond (ns) probe delay (t). Schematic description of excited-state CD is also included in the inset. (b) Calculated electron and hole natural transition orbitals (NTOs) for the excited-state transitions responsible for the observed excited-state CD in BNPB. Red arrows on the NTOs denote the significant electronic density on the binaphthyl group, suggesting that the excited-state CD in BNPB arises due to mixing between the molecular orbitals of pentacene and binaphthyl moieties.

As shown in Figure 4a, the shape of the calculated CD spectrum of BNPB agrees very well with the measured CD spectrum. Interestingly, the magnitude of the calculated CD is 20 times smaller than the magnitude of the measured CD of BNPB. This discrepancy is surprising given the conformational rigidity in BNPB enforced by the binaphthyl unit.³⁴ To investigate whether the observed discrepancy between the magnitude of the calculated and the measured CD is due to changes in the molecular geometry of BNPB, we calculate the CD for different geometries of BNPB by varying the dihedral angle (α) between the pentacene units and the torsional angle (β) between the pentacene and the binaphthyl unit (Figure 4b). In this regard, we utilize the dissymmetry parameter (g_{abs}), defined as the ratio of CD and absorbance (eq 4).⁵⁵

$$g_{\text{abs}} = \frac{2(A_L - A_R)}{(A_L + A_R)} \quad (4)$$

As shown in Figure 4c, the maximum value of g_{abs} calculated through excitonic coupling of monomeric transitions in BNPB is an order of magnitude smaller than the measured g_{abs} . Thus, the high magnitude of the measured CD cannot be accounted for by simply considering changes in the molecular geometry of BNPB with respect to the ground-state-optimized geometry, that is, the assumption that excitonic coupling involves purely “monomeric” transitions does not hold true for the transitions in the region 250–400 nm.

Further insights into the nature of transitions in BNPB can be obtained by probing the excited-state chiroptical response through excited-state CD measurements. With reference to the observed CD in BNPB solutions for the $S_0 \rightarrow S_n$ ($n \geq 3$) transitions in the UV region, we probe the excited-state CD in the wavelength region 1150 nm $< \lambda <$ 1450 nm, corresponding to the energy difference (~ 1 eV) between the S_1 state and the S_3 singlet excited state (see inset of Figure 5a). Excited-state CD of BNPB solution is obtained by calculating the difference

in the PIA of the L- and R-handed circular polarizations of the probe pulse (eq 5).

$$\begin{aligned} \text{excited-state CD}(t) &= \text{PIA}_L(t) - \text{PIA}_R(t) \\ &= \log \left(1 + \left(\frac{\Delta T}{T} \right)_L(t) \right) - \log \left(1 + \left(\frac{\Delta T}{T} \right)_R(t) \right) \end{aligned} \quad (5)$$

Here, t refers to the time delay between the pump and the probe pulse and $\text{PIA}_L(t)$ and $\text{PIA}_R(t)$ represent the PIA at probe delay t when the probe pulse polarization is L and R, respectively, where $\left(\frac{\Delta T}{T} \right)_L(t)$ and $\left(\frac{\Delta T}{T} \right)_R(t)$ are the differential changes in the transmission intensity at probe delay t for L and R polarizations of the probe, respectively.

In the wavelength region 1150 nm $< \lambda <$ 1450 nm, BNPB solutions in DCM exhibit a broad PIA feature which decays with a single-exponential lifetime of around 9 ns (Figure 5a). This decay is similar to the photoluminescence lifetime (~ 8 –9 ns) of BNPB in DCM, suggesting that the origin of PIA in the NIR region is due to absorption from the S_1 state. BNPB shows a significant bisignate excited-state CD response centered on the PIA maximum at 1380 nm attributed to the excited-state excitonic coupling between monomeric excited-state singlet transitions. It should be noted that the excited states of the binaphthyl moiety are significantly high in energy (>3.8 eV);⁵⁶ as such, excited-state CD in the NIR region (0.85–1.08 eV) cannot be explained by the direct excitation of the binaphthyl excited-state manifold from the S_1 (~ 1.9 eV) state of BNPB. The shape of the excited-state CD does not change in the measured time window (1–20 ns), suggesting no change takes place in the molecular electronic state and/or excited-state orientation during this time scale. Further, the decays of the excited-state CD and PIA are very similar,

suggesting that no change occurs in the relative orientation of the pentacene units.

To understand the origins of the excited-state CD and PIA response in **BNBP**, we turn to excited-state calculations (Figure 5b). We used quadratic response DFT (QR-DFT) to calculate the electronic transitions between excited states ($S_n \rightarrow S_m$) in **BNBP**. The transition dipole moments between two excited states n and m (with $n \neq m$) from a reference ground state S_0 is calculated from the double residue of the quadratic response function with dipole operators. We computed transitions between the eight lowest excited states (from S_1 to S_4) using the B3LYP exchange-correlation functional and a 6-31G* basis set. We also performed calculations using the long-range-corrected CAM-B3LYP and the hybrid BhandHLYP functional (Note S4). All QR-DFT calculations were performed using the DALTON software.^{57,58} In agreement with the experimental PIA spectrum, we find the excited-state transitions with nonzero oscillator strengths in the infrared between 1100 and 1500 nm (see Note S4, Figures S15–S22). To further obtain information on the chiroptical properties, we computed the rotatory strengths for transitions between the electronic excited states both in the length and in the velocity gauge using the excited-state matrix elements for electronic dipole, magnetic dipole, and velocity operators obtained as double residues of quadratic response functions.^{59,60} The simulated excited-state CD spectrum (see Figures S15–S22, Table S4) reproduces the shape and sign of the measured excited-state CD spectrum when considering transitions from the lowest excited S_1 state ($S_1 \rightarrow S_3$ and $S_1 \rightarrow S_4$).

To gain information on the localization pattern of the excited states involved in the NIR PIA feature, we calculated the hole and electron natural transition orbitals (NTOs) of the first 4 singlet excited states using time-dependent density functional theory (TD-DFT) on the basis of ground-state-optimized geometry of **BNBP**. Given the molecular dimer structure, the energy states are double degenerate and appear in pairs. NTOs were obtained using the Gaussian 16 software⁶¹ using a range of exchange-correlation functionals, namely, B3LYP, CAM-B3LYP, BHandHLYP, PBE1PBE, LC-wPBE, and wB97XD with a 6-31G* basis set. The results regarding the NTO localization pattern are similar across the different functionals, and in Figure 5b we display the B3LYP-computed NTOs for the excited states relevant to the excited-state CD response.

As expected, the electron and hole NTOs of the S_1 state of **BNBP** are localized on the pentacene units, revealing that the lowest energy singlet excitation in **BNBP** is monomeric in nature (Figure 5b). Surprisingly, the electron densities of the states responsible for the observed CD and excited-state CD response are shared between the pentacene units and the binaphthyl moiety. This suggests that excited-state CD in **BNBP** is induced by the binaphthyl bridge and the electronic density on the bridge plays a significant role in determining the rotatory strength.

Thus, utilizing the CD and excited-state CD responses, we conclude that $S_0 \rightarrow S_3$ and other higher energy singlet transitions in **BNBP** involve significant delocalization of the electronic density on the “bridging binaphthyl group”. As a result of the involvement of the chiral bridge, these transitions exhibit large rotatory strength resulting in significant CD and excited-state CD response in the UV and NIR regions, respectively. Comparing the dynamics of iSF in **BNBP** to a number of other iSF-based molecular geometries reported in

the literature (Table S2), we observe that **BNBP** uniquely affords a quantitative decoupling of the TT state into spin-uncoupled triplets $T_1 + T_1$. Given the recent reports^{26,27} on the significant role of the high-energy molecular orbitals of the bridge on the spin decoupling dynamics of the TT state, it is likely that the observed delocalization of the electronic density on the binaphthyl moiety aids in the spin decoupling of the TT state into $T_1 + T_1$. Further investigations focusing on the delocalization of electron density into the bridge can impart insights into the processes that enable decoupling in the TT state. For example, techniques such as resonance Raman spectroscopy can be used to develop insights into the role of molecular vibrations that affect spin decoupling in the TT state.⁶² Furthermore, magnetic circular dichroism (MCD) spectroscopy, which probes the magnetic field-induced mixing between close-lying molecular orbitals,⁶³ can be used to investigate the extent of the pentacene–bridge wave function overlap and its effect on the coupling between the different pentacene units in such systems. We note that while **BNBP** affords a quantitative decoupling of the TT state, the rate of TT formation in **BNBP** is slow. This is likely due to a weak coupling between the pentacene units in **BNBP**.⁷

Although we do not observe any effect of molecular chirality on the nature of the spin of the states in **BNBP**, likely due to a weak coupling of the electron spin to the molecule, we anticipate that the design of chiral bridges that allow a significant coupling of the electron spin to the chiral molecular framework is an interesting direction for future work. Such molecules can exhibit chiral-induced spin selectivity (CISS),^{64,65} enabling a much desired spin control on the iSF process.

CONCLUSIONS

We utilized CD and excited-state CD measurements of chiral pentacene dimer derivative (**BNBP**) bearing an axially chiral binaphthyl bridge to demonstrate that the chiroptical response of the system can be used as a sensitive marker of monomer-bridge interactions in the ground and excited states. Specifically, the analysis of CD and excited-state CD signatures of isolated **BNBP** molecules in solution reveals that high energy singlet transitions are significantly perturbed by the binaphthyl bridge; this likely aids in the decoupling of the triplet pair into uncoupled triplets.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpca.1c05254>.

Experimental procedures; model for fitting excited-state dynamics; TAS measurements at different fluences and different solvents; details of the theoretical and computational analyses (PDF)

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